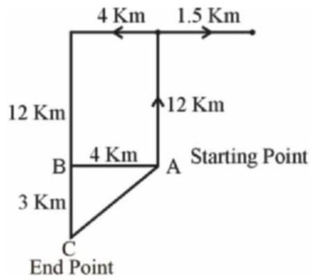
58. A person walks 12 km due north, then 15 km due east, after that 19 km due west and then 15 km due south. How far is he from the starting point?

(a) 5 km (b) 9 km
(c) 37km (d) 61 km

Ans. (a) :



Distance between starting point A and end point C is-

$(AC)^2 = (AB)^2 + (BC)^2$

$(AC)^2 = 4^2 + 3^2$

$(AC)^2 = 25 \Rightarrow AC = 5$

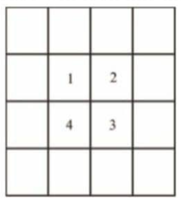
$\therefore$ Distance between starting point and end point in 5Km.

59. A cube has all its faces painted with different Colurs. It is cut into smaller cubes of equal sizes such that the side of the small cube is one-fourth the big cube. The number of small cubes with only one of the sides painted is :

(a) 32 (b) 24
(c) 16 (d) 8

Ans. (b) :

Each face of the cube will look like the figure below:



On each face, the smaller cubes which do not lie on the edges of bigger cube will have only one side painted.

As shown in the figure above, there are four cubes which has only on face painted. accordingly, there will be (6) such faces of bigger cube so total 24 small cubes have one painted face.

Hence 24 is correct answer.

60. Ram and Shyam work on a job together for four days and complete 60% of it. Ram takes leave then and Shyam works for eight more days to complete the job. How long would Ram take to complete the entire job alone ?

(a) 6 days (b) 8 days
(c) 10 days (d) 11 days

Ans. (c) : Let : R $\rightarrow$ Ram, S $\rightarrow$ Shyam

$\because \quad \frac{4}{R}+\frac{4}{S}=\frac{60}{100}$

$4\left[\frac{1}{R}+\frac{1}{S}\right]=\frac{60}{100}$

$4\left[\frac{1}{R}+\frac{1}{S}\right]=\frac{3}{5}$

$\frac{1}{R}+\frac{1}{S}=\frac{3}{20}$ .....(i)

According to the question,

$\frac{4}{R}+\frac{12}{S}=1$

$\frac{4}{R}+\frac{4}{S}+\frac{8}{S}=1$

$4\left[\frac{1}{R}+\frac{1}{S}\right]+\frac{8}{S}=1$

$4\times\frac{3}{20}+\frac{8}{S}=1$

$\frac{8}{S}=1-\frac{12}{20} \Rightarrow \frac{8}{S}=\frac{8}{20}$

$\frac{1}{S}=\frac{1}{20}$ .....(ii)

Putting values from equation (ii) to (i)-

$\frac{1}{R}+\frac{1}{20}=\frac{3}{20} \Rightarrow \frac{1}{R}=\frac{3}{20}-\frac{1}{20}$

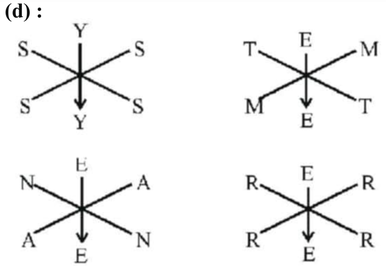
$\frac{1}{R}=\frac{2}{20} \Rightarrow \frac{1}{10}$

Therefore, time taken by Ram to complete the entire job alone = 10 days.

61. A military code writes SYSTEM AS SYSMET and NEARER as AENRER. Using the same code, FRACTION can be written as :

(a) CARFTION (b) FRACNOIT
(c) NOITCARF (d) CARFNOIT

Ans. (d) :



$\therefore$ Correct answer in (d) CARFNOIT

62. If R and S are different integers both divisible by 5, then which of the following is not necessarily true ?

(a) R – S, is divisible by 5
(b) R + S, is divisible by 10
(c) R × S, is divisible by 25
(d) $R^2 + S^2$, is divisible by 5

Ans. (b) :

According to question-

Let, R = 10 and S = 5

then, (a) $\frac{R-S}{5}=\frac{10-5}{5}$

= 1 Whole Number

(b) $\frac{R+S}{10}=\frac{10+5}{10}=\frac{15}{10}$

= 1.5 not a Whole Number

(c) $\frac{R\times S}{25}=\frac{10\times 5}{25}=\frac{50}{25}$

= 2 Whole Number

(d) $\frac{R^2+S^2}{S}=\frac{(10)^2+(5)^2}{5}=\frac{100+25}{5}=\frac{125}{5}$

= 25 Whole Number

Therefore, $\frac{R+S}{10}$ is not necessarily true. The correct answer is option (b).

63. How many numbers are there between 100 and 300 which either begin with or end with 2 ?

(a) 110
(b) 111
(c) 112
(d) None of the above

Ans. (a) :

Number between 100 and 200 which either begin with or end with 2 are 102, 112, 122, 132, 142, 152, 162, 172, 182, 192 = 10

between 200 to 300 are 100

$\therefore$ Total = 100 + 10 = 110.

Directions for the following 8 (eight) items : Read the following five passages and answer the items that follows each passage. Your answers to these items should be based on the passages only .

Passage-1

As we look to 2050, when we will need to feed two billion more people, the question of which diet is best has taken on new urgency. The foods we choose to eat in the coming decades will have dramatic ramifications for the planet. Simply put, a diet that revolves around meat and dairy a way of eating that is on the rise throughout the developing world, will take a greater toll on the world's resources than one that revolves around unrefined grains, nuts, fruits and vegetables.

64. What is the critical message conveyed by the above passage?

(a) Our increasing1 demand for foods sourced from animals puts a greater burden on our natural resources
(b) Diets based on grains, nuts, fruits and vegetables are best suited for health in developing countries
(c) Human beings change their food habits from time to time irrespective of the health concerns
(d) From a global perspective, we still do not know which type of diet is best for us

Ans. (a) : The conclusive message that emerges from the above passage in that the increasing demand for animal based food sources places a greater burden on our natural resources. A diet high in meat and dairy products will have an impact on human food production.

Passage 2

All humans digest mother's milk as infant, but until cattle began being domesticated 10,000 years ago, Children once weaned no logger needed to digest milk. As a result, they stopped making the enzyme lactase, which breaks down the sugar lactose into simple sugars. After humans began herding cattle, it became tremendously advantageous to digest milk, and lactose tolerance evolved independently among cattle herders in Europe, the middle East and Africa. Groups not dependant on cattle, such as the Chinese and Thai, Remain lactose intolerant.

65. Which among the following is the most logical assumption that can be made from the above passage ?

(a) About 10,000 years ago, the domestication of animals took place in some parts of the world
(b) A permanent change in the food habits of a community can bring about a genetic change in its members
(c) Lactose tolerant people only are capable of getting simple sugars in their bodies
(d) People who are not lactose tolerant cannot digest and dairy product

Ans. (b) : The most logical assumption that can be made from the above passage is that a permanent change in the eating habits in a community can bring about genetic changes in the members of the community.

Passage 3

"The conceptual difficulties in National Income comparisons between underdeveloped and industrialized countries are particularly serious because a part of the national output in various underdeveloped countries is produced without passing through the commercial channels.''

66. In the above statement, the author implies that :

(a) the entire national output produced ad consumed in industrialized countries passes through commercial channels
(b) the existence of a non-commercialized sector in different underdeveloped countries renders the national income comparisons over countries difficult